Reddy Rewat

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.89
TIRUMALA JUNIOR COLLEGE	Class XII (Andhra Pradesh Board of Intermediate Education (BIEAP))	2021	80.30%
DR K K R'S GOWTHAM HIGH SCHOOL	Class X (Board of Secondary Education, Andhra Pradesh (BSEAP))	2019	9.30%

SKILLS

Programming: Python, C++, SQL

AI/ML: Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn, NLP

Data Analytics: Pandas, NumPy, Power BI, Tableau, Time-Series Analysis

Energy Systems: Smart Grid, Microgrid, Solar Forecasting, Energy Analytics

Development: FastAPI, React.js, Node.js, Git, GitHub

RELEVANT COURSES

Data Science Analysis | Smart Grid | Energy Management | PV Technology | Power Converter Control | Energy Storage | Renewable Energy Systems | Electric Vehicles | Computational Modelling

PROJECTS

M.Tech Thesis: 10 kW AI-Enabled Smart Dual-Microgrid with Digital Twin

- Developing an AI/ML-driven framework for smart grid and microgrid energy management, integrating time-series forecasting, predictive analytics, anomaly detection, optimization, real-time monitoring, and data visualization. Applying machine learning approaches for renewable generation forecasting and energy prediction under dynamic operating conditions to support intelligent and data-driven energy management.

AI-Agentic Smart Microgrid Monitoring and Resilience Framework

- Developed an AI-agentic microgrid simulation and monitoring framework for intelligent grid-state analysis, anomaly response, and resilient energy-system operation. Modeled an 8-node and 11-edge interconnected network for power-flow and grid-disturbance simulation, integrating Python, FastAPI, WebSockets, multi-agent AI workflows, and interactive 2D/3D visualization for real-time monitoring and analytics.

Multi-Horizon Solar Power Forecasting using CNN-Transformer

- Developed a CNN-Transformer deep learning architecture for multi-horizon photovoltaic power forecasting from +1 hour to +1 day. Performed time-series preprocessing, feature engineering, temporal pattern extraction, and sequence modelling to capture short- and long-term dependencies in solar generation data, and evaluated forecasting performance against baseline models to improve prediction reliability.

Deep Learning-Based Synthetic Media Detection and Forensic Analysis

- Developed a CNN-based computer vision pipeline for deepfake and manipulated-media detection, achieving approximately 93% accuracy on a benchmark dataset. Implemented face detection, frame extraction, preprocessing, feature extraction, and CNN-based classification using OpenCV for automated synthetic-media analysis.

PUBLICATIONS

Paper titled "**Optimizing Stroke Detection with Genetic Algorithm-Based Feature Selection in Deep Learning Models**" published in *Applied Neuropsychology: Adult* (Taylor & Francis), 2025.DOI: 10.1080/23279095.2025.2516259

WORK EXPERIENCE & INTERNSHIPS

Certified Trainee (Technical) | Tech Mahindra

- Completed technical training in SQL, HTML, CSS, JavaScript, and Full-Stack Development, with exposure to software development practices, debugging, code quality, and application development.

Full Stack Developer | Kalam Dream Labs

- Developed full-stack web applications across frontend and backend components, implementing responsive interfaces, API integrations, and application functionality using JavaScript-based technologies.

AWS Data Engineering Intern | EduSkills Foundation

- Worked on cloud ETL pipelines using AWS Glue, S3, and Redshift, with real-time streaming through Kinesis and analytics using QuickSight; explored AWS EMR for big-data processing.

Machine Learning Intern | Jobiak

- Worked on Machine Learning and NLP for job-posting analysis, performing data preprocessing, NLP model development, feature processing, and model optimization using Python.

AI/ML Virtual Internship | Google

- Completed project-based training in AI/ML, covering ML workflows, model development, data processing, and practical AI applications.

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

5th Position, Beacon Technologies AI Foundry'26 -- State level Hackathon, 2026

10th Position, Devshouse'24 -- Nationwide Hackathon organized by GDSC, 2024

OCS Student Volunteer, IIT Hyderabad -- Jul 2025-Present